

Executive Summary:

Road Accident Dashboard (2021 vs 2022)

The **Road Accident Dashboard** provides an in-depth comparative analysis of road accident data between the **Current Year (CY – 2022)** and the **Previous Year (PY – 2021)**. Across all major metrics — including **Total Casualties, Total Accidents, and Fatal Casualties** — the analysis reveals a **significant year-over-year (YoY) decline**, signaling measurable improvement in overall road safety.

Key Highlights

- **Total Casualties:** ↓ ~6.8% to 11.9% YoY
- **Total Accidents:** ↓ ~6.5% to 11.7% YoY
- **Fatal Casualties:** ↓ ~17.8% to 33.3% YoY

While this decline is a positive sign, **Slight Casualties** continue to form the bulk of total incidents (~84.1%), and **Single Carriageway roads** remain the most accident-prone. The dashboard visualizations effectively highlight these trends, helping policymakers identify high-risk categories by **severity, road type, time of day, and vehicle involvement**.

Key Findings from the Data

1 Overall Trend is Positive (Decline)

All major indicators show consistent YoY decreases, confirming a general improvement in road safety.

- **Example:** CY Total Casualties ≈ 195.7K (–11.9% YoY)
- **Example:** CY Fatal Casualties ≈ 2.9K (–33.3% YoY)

2 Severity Mix is Highly Skewed

- **Slight Casualties:** Represent ~84.1% of total casualties (e.g., 351,436 of 417,883).
- **Fatal Casualties:** Though only ≈1.7%, their sharp reduction (>30%) is a major success.

3 Key Causal Factors and Locations

- **Road Type:** *Single Carriageways* dominate, with ≈309.7K casualties in one dashboard view.
- **Urban vs. Rural:** Urban areas (≈61.95%) exceed rural areas (≈38.05%) in casualty share.
- **Time of Day:** Most casualties occur **during daylight** (≈74%), but night-time severity remains a concern.
- **Vehicle Type:** Cars are involved in the majority (≈155.8K, ~79.8%), followed by bikes/motorcycles (~15.6K).

Monthly Trend (Seasonality)

Casualties show a recurring seasonal pattern, peaking during **late summer to autumn (August–October)** in both years — a potential focus period for safety campaigns.

Suggestions & Recommendations

A. Actionable Safety Interventions

- **Target Single Carriageways:** Prioritize safety audits, maintenance, and speed enforcement on these routes.
- **Focus on Slight Casualties:** Develop awareness campaigns around minor accidents and driver attentiveness.
- **Vehicle-Specific Programs:**
 - Cars: Promote defensive driving and refresher courses.
 - Bikes: Encourage visibility and safe riding campaigns.

B. Dashboard & Analytical Enhancements

- **Standardize KPI Timeframes:** Clearly label CY and PY as “2022” and “2021” throughout visuals to avoid confusion.
- **Severity in Monthly Trends:** Add filters for *Fatal*, *Serious*, and *Slight* categories in monthly trend charts.
- **Include Normalized Metrics:** Display casualties per 1,000 vehicles or per million miles driven (if data available).
- **Visualize “After Accident” KPI:** Introduce a single KPI card (e.g., gauge or number card) showing total casualties post-accident, benchmarked against targets or totals.

In summary, while the year-over-year reduction in total and fatal casualties marks a substantial achievement, the concentration of minor accidents on **Single Carriageways** and during **daytime hours** highlights areas for continued safety intervention.

Future dashboard enhancements can provide even deeper insights through improved KPI normalization and severity-level segmentation.